Mark Menezes

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Education

Texas A&M University College Station, TX
BS, Mechanical Engineering, 3.4 GPA *August 2023 – May 2027*

Experience

Canon Nanotechnologies
Mechanical Engineering Intern Austin, TX
Nano-inkjet Lithography Group *May 2026 – August 2026*

- Conceptualized, designed, and manufactured testbeds for fatigue testing of vacuum components for the wafer lift mechanism, condensing the testing cycle from one year to two weeks
- Developed software for the testbed in C++ to gather data from vacuum sensors, a stepper motor, and a bluetooth module to wirelessly transmit over WiFi to enable live data analysis
- Created custom fittings for peristaltic pumps endurance testing, with components engineered for a two-year testing life cycle
- Designed and assembled an enclosure to test resist curing under high-power UV LEDs, increasing operator safety, and reducing sample testing iteration time

Texas A&M University Robotics Automation and Design (RAD) Lab
Robotics Intern College Station, TX
Manipulator Group *May 2025 – August 2025*

- Designed, produced, and tested a modular parasitic torque testbed to measure resistance created by externally mounted vendor cable harnesses and PTFE fluid hoses on a robotic arm designed for LEO
- Modeled and manufactured an enclosure to test custom motor controllers with an absolute rotary encoder, achieving perfect concentricity and z-height
- Developed 3D printed mounting components for custom PCB's and 1:1 models of robotic actuator components to visualize assembly and manufacturing

Space Modular Manipulator Group *May 2024 – August 2024*

- Designed robotic actuators for a 7-DOF robotic arm, reducing mass and improving harmonic drive through-bore efficiency while implementing SEA geometries for torque calculations
- Developed a wire harness fatigue testbed, simulating a space environment, modeling a vacuum, and heating/cooling, for life-cycle validation over 1 million cycles
- Created engineering calculators in Visual Basic to determine minimum bolt sizes for dynamic fixtures, enhancing safety and efficiency in structural design

Projects

Engineer, Formula SAE Electric *September 2023 – Present*

- Led a team of four engineers to design a rigid and lightweight FSAE chassis, completing Python tube optimization, structural equivalency calculations, and torsional deflection simulations. Cutting subsystem mass by 7.47%
- Automated a previously manual data conversion from the simulation software (Optimum Kinematics) to the modeling software, decreasing task completion time by 80% and removing human error
- Led end-to-end design and fabrication of a custom Pro-Ackermann steering system, optimizing lateral force and handling characteristics while minimizing tire scrub
- Designed and manufactured a custom carbon fiber integrated seat, headrest, and firewall to reduce mass and improve ergonomics with driver-specific molded inserts

Technical Skills

Software: SolidWorks, NX, PTC Creo, ANSYS, Optimum Kinematics, Multisim, Python, C++
Manufacturing: 3D Printing, Mill, Lathe, CNC, TIG Welding
Languages: English, Spanish, Portuguese
Security: US Citizen, Controlled Unclassified Information (CUI)
Hobbies: Figure Skating, Soccer, Beekeeping